CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (EC)(C.S.E.) Sem. - I University Examination, 2010-11
EC- 705 Advanced Digital Satellite Communications

Date: 13/05/2011, Friday **Time: 10:00 a.m to 1:00 p.m**
Maximum Marks: 70

Instructions:
(1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work is done in the last page of main answer book; don't write anything on the question paper.

SECTION- I

Q-1 (a) Define & discuss the ***Keplerian elements*** in brief. **2**

(b) Describe the characteristics of geostationary satellite. **2**

(c) A satellite link operating at 14 GHz has receiver feeder losses of 1.5dB and a free space loss of 207 dB. The atmospheric absorption loss is 0.5dB and the antenna pointing loss is 0.5dB.Depolarization losses may be neglected. Calculate the total link loss for clear sky conditions **3**

(d) Define the following terms: **4**
Apogee, prograde orbit, MUF, Critical frequency

Q 2 (a) With the help of neat diagram, justify the needs for TT & C as a subsystem for a successful satellite communication system. **5**

(b) Determine the azimuth angle for a geostationary satellite at 90 degrees W, for an earth station at latitude 35 degree N, longitude 100 degrees W. **4**

(c) In a satellite circuit the carrier to noise spectral density ratios are: uplink 23dBHz, downlink 20 dBHz, intermodulation 24dBHz. Calculate the overall carrier to noise ratio. **3**

OR

Q 2 (a) Find the time in Julian Centuries from the reference time January 0.5,1900 to13h UT on 18 December 2000. **2**

(b) Discuss the ***basic elements of redundant earth station*** with the help of neat block diagram. **4**

(c) A satellite is orbiting in the equatorial plane with a period from perigee to perigee of 12hr.Given that the eccentricity is 0.002.Calculate the semi major axis. The earth equatorial radius is 6378.1414km.(μ=3.986005*10^{14} m^3/cm^2) **4**

(d) Determine the orbit eccentricity of a satellite moving in elliptical path having semi major axis equal to 16000 km if the difference between perigee and apogee is 25000 km and earth radius is 6360 km. **2**

Q 3 (a) Define ***saturation flux density*** and prove that **6**

$[EIRP_S]_U=[\Psi_S]+[A_0]+[LOSSES]_U-[RFL]$

(b) A circularly polarized wave at frequency of 12GHz is transmitted from satellite. The point rain rate is 13mm/h.calculate the specific attenuation. (a_h=0.0188, a_v=0.168,b_h=1.217,b_v=1.2 for f=12GHz) **2**

(c) Explain how depolarization is caused by rain. How does it differ from depolarization caused by ice? **4**

OR

Q 3 (a) Given that L_U=200db, L_D=196db, G_E=G'_E=25db, G_S=G'_S=9db, G_{TE}=G_{RE}=48db, G_{RS}=G_{TS}=19db, U_S=U'_S=1μJ, and U_E=U'_E=10 μJ, Calculate the transmission gain[γ], interference level [I_1] and [I_2], and the equivalent temperature rise overall. **5**

(b) Distinguish between the ***cross polarization discrimination*** and the ***polarization isolation*** phenomena. **4**

(c) Define ***effective isotropic radiated power***. A transmitter feeds a power of 10W into an antenna which has a gain of 46 db. Calculate the EIRP in i) Watts ii) dBW. **3**

SECTION- II

Q 4 (a) Enlist various frequency bands used in satellite system. **1**

(b) What do you mean by ***energy dispersal***? **2**

(c) Compare the uplink power requirement for Time Division Multiple Access and Frequency Division Multiple Access techniques. **3**

(d) What do you mean by ***three axis stabilization*** of satellite body? **2**

(e) Enlist the physical and orbital characteristics of advanced Tirous - N space craft. **3**

Q 5 (a) Explain what is meant by satellite attitude and brief describe two forms of attitude control. **5**

	(b)	Describe the channeling scheme for SPADE system for Intelsat satellite. Discuss its DASS based communication system.	5
	(c)	Define the terms ***connect clip*** and ***freeze out*** used in connection with digital speech interpolation technique used for satellite communication.	2
		OR	
Q 5	(a)	Explain various causes and effects of transmission losses occurring in satellite communication system.	5
	(b)	Describe the principle of operation of Code Division Multiple Access. What effects do the unwanted signals have on the wanted signal?	4
	(c)	Write the technical note on mobile satellite system and the services offered by it.	3
Q 6		Attempt any three	12
	(a)	Describe and compare the master antenna TV system and community antenna TV system	
	(b)	Describe the operation of typical VSAT system.	
	(c)	Describe main feathers and services offered by the IRIDIUM satellite system.	
	(d)	Explain why a minimum of four satellites must be visible at any earth location utilizing the GPS system for position determination.	
	(e)	Draw and explain Receive-Only Home TV systems.	